

Stage A: prepare data once

2,000
questions

3 teachers
× 3 samples

18,000
candidates

answer
filtering

Qual(t)
embed(t)

filtered candidates (Fit reads every chain)

cached Qual(t), embed(t), reused by every method

Stage B: per student × per method

Fit(t, m)
once per student

exact search
max $F(S, m)$

5,670 samples
k = 3 per question

QLoRA
fine-tuning

next method: reselect and retrain

next student: recompute Fit, then repeat every method

fine-tuned LoRA adapter, one per student × method × seed

Stage C: evaluate and deploy

6 benchmarks
Acc@4, Acc@1

GGUF
Ollama